SUBHAM

Software Developer | Advanced Manufacturing and Design (AMD) - Robotics | IIT Jodhpur

@ subhamk356@gmail.com (+91) 8533027637 linkedin.com/in/upadhayay Github Leetcode Noida, India

EXPERIENCE

Lead Software Engineer | Bitsilica -> LG Soft India | 03/2026 - Present

Bengaluru, Karnataka

- Worked on LG's Robot Vacuum Cleaner Software Stack (C++).
- Handled navigation and cleaning modules.
- Developed and owned modules like carpet cleaning, steam cleaning, etc.
- Got Appreciation for the robot vacuum cleaner project. See Certificate
- **Technologies Used:** Navigation and SLAM, C++, Data Structures, OOPs, Git, CMake, Finite State Machines

Lead Robotics Software Engineer | Addverb | 05/2024 - 11/2025

Noida, Uttar Pradesh

- Worked on the Fleet Management (C++) of **Sorting Robots Fleet** and path planning (A*, Dijkstra etc).
- Developed and performed simulations to achieve throughput for various clients.
- Developed various feature to improve and analysis the fleet of mobile robots, such as heatmap visualization for errors, drops, and pickings.
- Spent time on site at Mercado Livre Brazil, responsible for the FMS and Robot feature development and testing different modules.
- **Technologies Used:** C++, Data Structures, OOPs, ROS, Git, CMake, APIs, MongoDB, MQTT, Docker, Kubernetes, and Grafana Dashboards, Behavior Trees (BTs)

Software Developer | Nestbotics | 12/2023 - 05/2024

Noida, Uttar Pradesh

- Worked on Path planning of Multiple agents.
- Worked on Fleet Management of **AS/RS** : Write Finite State Machine to handle Multiple agents.
- **Traffic Manager**: Make Traffic manager which is responsible to handle collision conditions and deadlocks.
- **Task Scheduler**: Assigning Tasks to agents in efficient manner.
- Worked on ROS 2 packages, SLAM and Gazebo.
- **Technologies Used:** C++, Data Structures, OOPs, Git, ROS 2, Gazebo, SLAM, CMake, APIs, SQL, MQTT

Software Developer (R&D) | Falcon AutoTech | 08/2021 - 11/2023

Noida, Uttar Pradesh

- Worked in R&D IT department for the project of **3D Shuttles AS/RS** System (NEO).
- Worked on **path planning for multiple robots** in an ASRS System. Used the A-Star algorithm for 3D path planning.
- Presented The ASRS system in India Warehousing Show (**IWS 2021 & 2023**).
- Spent time on the customer site while installing ASRS in Delhivery's and then the Decathlon Warehouse.
- Got award of **Employee of the Month** for my contribution in Neo ASRS project. See Certificate
- **Technologies Used:** C#, Git, SQL, ASP.Net MVC, Data Structures, OOPs, HTML, CSS, JavaScript, C++

PROJECTS

Modelling and Control of Multi Mobile Robots | Github

06/2020 - 07/2021

Guides – Dr. Suril V. Shah and Dr. Niladri S. Tripathy | IIT Jodhpur

- Used **Model Predictive Control** for trajectory generation.
- **Motion Planning** For both **Omnidirectional** and **Non-Holonomic** robots
- Performed **Online Optimization** for generating trajectories in **real time**.
- Used **On-demand Collision Avoidance** method to avoid collision among multi agents.

Segregation of Heterogeneous Robotic Swarms | Github

12/2020 - 02/2021

Guide – Dr Anoop Jain, Department of Electrical Engineering IIT Jodhpur.

- Applied the concept of **Artificial Potential Fields** to Segregate a swarm.
- Asymptotically Flocking of the system while segregating.

Path Planning of RRR arm using Potential Field | Github

May – June 2020

Guide – Dr. Suril V. Shah, Department of Mechanical Engineering IIT Jodhpur.

- Using MATLAB performed **path planning** of the RRR arm from initial to final configurations such that it avoids collisions with **static obstacles** and **self-collision** of the links.

Generalized Inverse Kinematics of 6-DOF PUMA Robot | Github

Feb – March 2020

Guide – Dr. Suril V. Shah, Department of Mechanical Engineering IIT Jodhpur.

- Find all possible solution of a **6-DOF PUMA Robot** for a given position and orientation of end-effector by performing inverse kinematics using **kinematic decoupling**.

Path Planning Using Rapidly Exploring Random Trees | Github

- Created a user interface using tkinter module.
- Worked on **RRT**, **RRT Star** and **Biased RRT**.

CERTIFICATION AND ONLINE COURSES

- Machine Learning A-Z [2026]: Python | Udemy | See credential
- Advanced Kalman Filtering and Sensor Fusion | Udemy | See credential
- Introduction to Self-Driving Cars | Coursera | See credential
- Mechanics and Control of Robotic Manipulators | NPTEL | See credential
- Robotics: Aerial Robotics | Coursera | See credential
- **ROS2** For Beginners (ROS Foxy - 2021) | Udemy | See credential
- Algorithmic Toolbox | Coursera | See credential
- Localization, Navigation and SLAM | Udemy | See credential
- Robot Operating System | Udemy | See credential

EDUCATION

M.Tech in AMD - Robotics
2019 – 2021 CGPA : 8.41

Indian Institute Of Technology (IIT), Jodhpur

B.Tech in Mechanical Engineering
2014 – 2018 CGPA : 8.0 (76.2%)

Dr. APJ Abdul Kalam Technical University

Intermediate (HSC)
2012 - 2014 Per : 80.00%

S.D. Inter College, Saharanpur

Matriculation(SSC)
June 2012 Per : 82.50%

R.C. Inter College, Saharanpur

SKILLS

Programming Languages:	C++ C# Python MATLAB Bash LaTeX HTML CSS JavaScript
Databases:	MySQL MongoDB
Robotics Frameworks:	ROS/ROS2 Gazebo CoppeliaSim rviz
Algorithms & Concepts:	OOPs Data Structures & Algorithms Path Planning (A*, Dijkstra, RRT) SLAM
Tools & Platforms:	Linux .net Git Gitlab Bitbucket Raspberry Pi Behavior Trees (BTs)

RELEVANT COURSES

Robotics | Advanced Control Systems | Networked Dynamical Systems and Control | Swarm Robotics | Artificial Intelligence-I | Machine Learning-I: Supervised Learning | Geometric Modelling and CAD | Theory of Machines

INTERNSHIPS AND TRAININGS

Laboratory Assistant | Jan – March 2020

IIT Jodhpur

- Through my work in the laboratory, I have refined my skills in writing reports, designing high-impact PowerPoint presentations, and team leadership.

Summer Internship | July – Aug 2017

Roadways Bus workshop, Saharanpur

- In the workshop, I learned about Gears and Gearboxes used in buses.

EXTRA CURRICULARS

- Joined The National Service Scheme (**NSS**) for two years.
- Earned a certificate in **Documentation Assistant** offered by PMKVY.
- Gave group dance performance in several events.

HOBBIES

Dancing | Photos and Videos Editing | Playing Table Tennis | Watching Movies